



OPEN Artificial intelligence technology for music teaching reform mode under DCNN algorithm

Chang Liu, Ningning Shi✉ & Shen Jiang

The purpose of this paper is to explore the reform and innovation mode of music teaching driven by artificial intelligence (AI). By introducing the Dilated Convolutional Neural Network (DCNN) algorithm and further integrating the attention mechanism, an audio recognition model based on a Multi-Branch Fusion Network Based on Dilated Convolution and Attention Mechanism (MBFN-DCAM) is constructed to improve the accuracy and efficiency of music audio recognition. Horizontal comparison experiments show that MBFN-DCAM outperforms 6 representative state-of-the-art (SOTA) models, including Audio Spectrogram Transformer (AST) and ResNet-50, with a recognition accuracy of $95.65\% \pm 0.35\%$ ($p < 0.01$). Validated by a randomized controlled trial ($n = 60$), feedback provided by the model significantly improves students' pitch accuracy and enhances their Music Self-Efficacy Scale (MES). Furthermore, the inference latency of the model on Jetson Nano is only 82.4 ms, fully demonstrating its deployment advantages in resource-constrained environments. This paper provides an efficient, robust, and educationally effective technical approach for intelligent music teaching evaluation.

Keywords Music teaching mode, Artificial intelligence, Dilated convolutional neural network, Audio recognition, Deep learning

Research background and motivations

In recent years, with the rapid development of artificial intelligence (AI) technology, especially the successful application of deep learning in many fields, the traditional education model is facing an unprecedented change. In music teaching field, despite the gradual infiltration of information technology, the teaching mode still depends on teachers' subjective judgment and experience accumulation, which is inefficient and difficult to personalize facing diverse student needs^{1,2}. However, the Convolutional Neural Network (CNN) in AI provides the possibility for the intelligent development of music teaching with its powerful feature extraction and pattern recognition capabilities³. Therefore, exploring AI technology, especially the application of CNN in music teaching, has become an important topic in current academic research and education reform.

The introduction of CNN changes the traditional music teaching method, and shows significant advantages in personalized teaching and large-scale education environment. Through CNN, the teaching system can automatically analyze a large amount of music data, accurately evaluate students' performance level, provide personalized learning feedback, and even dynamically adjust the teaching plan according to students' progress^{4,5}. Therefore, the CNN algorithm in AI technology has laid a solid technical foundation for becoming an important driving force of music teaching reform and realizing a more efficient and personalized teaching mode.

However, when dealing with 10-second music teaching audio, existing algorithms often lack systematic modeling of the long-range dependencies of temporal features. To strengthen the methodological foundation of this paper, it refers to the reasoning logic of advanced algorithms in complex signal processing. For example, Bai et al. (2025)⁶ demonstrated via an efficient bio-inspired engineering optimization algorithm that systematic architecture search optimization was critical for improving model robustness when handling high-dimensional data. Meanwhile, the improved parallel one-dimensional convolutional neural network proposed by Bai et al. (2025)⁷ for rolling bearing fault diagnosis showed the significant advantages of multi-channel feature extraction in capturing subtle changes in complex acoustic signals. Drawing on the argumentation style of these studies, this paper introduces a multi-branch architecture and dilated convolution into the music teaching scenario to address the problems of incomplete music feature capture and single evaluation dimension in traditional models.

College of Arts, Heilongjiang University, Harbin 150080, China. ✉email: 2003248@hlju.edu.cn

Research objectives

Most current deep learning-based research on music teaching evaluation is limited to technical recognition accuracy (e.g., consistency coefficients between AI and human teacher scores), often ignoring how algorithm outputs can be directly translated into improved student learning outcomes.

This paper focuses on the performance of the Multi-Branch Fusion Network Based on Dilated Convolution and Attention Mechanism (MBFN-DCAM) model on general datasets such as AudioSet, and attempts to explore the internal logic among model feedback, Music Self-Efficacy Scale (MES), and quantitative assessment of emotional expression. By establishing a closed loop from acoustic feature recognition to teaching intervention strategies, this paper aims to demonstrate that the introduction of AI technology can go beyond pure metric calculation and truly achieve substantial improvements in students' pitch correction and practice duration.

Compared with traditional models, the incremental contributions of this paper are mainly reflected in the following three aspects:

- Adopt a multi-branch fusion architecture to overcome the limitation of single convolution kernels in extracting multiple timbre features of music.
- Introduce the Dilated Convolution and Attention Mechanism (DCAM), which significantly improves the model's discrimination accuracy for key acoustic indicators in noisy teaching environments, rather than merely relying on basic feature superposition.
- Horizontal comparisons demonstrate that the model significantly reduces resource consumption while achieving recognition performance comparable to state-of-the-art (SOTA) models, enabling subsequent hardware implementation on edge devices.

Literature review

Review on the reform of music teaching mode

Driven by AI technology, music teaching is undergoing a profound transformation from the traditional master-apprentice system to a digital evaluation system. Zhang et al. (2024)⁸ discussed the problems existing in traditional basic music theory teaching in the information age and pointed out that using nonlinear analysis methods to deepen theoretical teaching remained a current challenge. Pavlenko et al. (2024)⁹ studied the role of modern computer technology in cultivating the professional abilities of music teachers and proposed a personalized education environment model. Liu et al. (2024)¹⁰ explored the application of mobile learning technology in online piano education and its impact on learning outcomes.

Although the above studies emphasize the construction of hardware environments, they remain weak in specific teaching intervention logic. Huang et al. (2023)¹¹ investigated the impact of AI-driven personalized recommendation systems on learner engagement in flipped classrooms. Yang et al. (2023)¹² explored how personalized learning analysis interventions enhanced learners' sense of achievement in blended learning. Critical analysis indicates that existing studies mostly focus on macro-level paradigm transformation while neglecting the refined design of underlying algorithm architectures in music signal processing. By designing a Multi-Branch Fusion Network (MBFN), this paper attempts to address the limitation of insufficient technical support precision in current teaching modes from the underlying algorithm.

The scene application of artificial intelligence in music education

AI technology offers the possibility of automated assessment and feedback for music education. Li et al. (2024)¹³ discussed the general trends in the application of AI technology in music education, while Yu et al. (2023)¹⁴ further summarized the contextual development of AI in music education. Regarding specific teaching assistance design, Dai et al. (2021)¹⁵ investigated AI-assisted instructional design schemes. For heterogeneous environments, Yang et al. (2023)¹⁶ explored a multi-source heterogeneous online education mechanism to optimize resource allocation.

However, the core challenge remains how to effectively translate technical recognition results into teachable strategies. Qian et al. (2023)¹⁷ discussed the role of human-centered design in improving the experience of AI information systems. Song et al. (2024)¹⁸ investigated the application of AI-assisted computing in automated scoring for piano education. Although these studies achieved basic evaluation functions, they generally lack a clear mapping logic between model outputs and practical teaching interventions. To this end, the proposed model MBFN-DCAM is specially designed with a feedback mapping mechanism, which aims to convert abstract recognition probabilities into actionable teaching strategies (e.g., pitch correction suggestions), thereby enhancing the practical guidance value of the technology.

Audio classification algorithm and advanced deep learning model

At the core algorithm level, audio recognition is evolving from traditional Convolutional Neural Network (CNN) toward more sophisticated architectures. Jena et al. (2023)¹⁹ investigated hybrid deep learning methods using wavelet transform and spectrogram analysis. Prabhakar et al. (2023)²⁰ explored efficient transfer learning for audio classification. Xie et al. (2024)²¹ studied models based on residual-gated CNN and attention mechanisms, demonstrating the advantages of attention in feature selection. For state-of-the-art (SOTA) models, Schmid et al. (2024)²² discussed the efficiency of dynamic convolutions as pre-trained audio models.

To further improve fine-grained performance, Tami et al. (2024)²³ investigated the performance of Transformer-based architectures in spectrogram processing. Bharti et al. (2025)²⁴ employed the Audio Spectrogram Transformer (AST) for voice activity detection and classification. Nguyen-Duc et al. (2025)²⁵ conducted a comparative study of spectrogram-driven deep audio models. Critically, despite their high accuracy, Transformers and AST incur substantial computational costs, which limit real-time deployment in real classrooms. This paper adopts Dilated Convolutional Neural Networks (DCNN) combined with the

Dilated Convolution and Attention Mechanism (DCAM). While maintaining the ability to capture long-term dependencies, the proposed method significantly reduces the number of parameters and inference latency, achieving a favorable balance between accuracy and deployment feasibility.

Emotional dimension of music education and summary of research blanks

To enhance the integrity of education, AI models must be capable of understanding emotion. Wang et al. (2023)²⁶ proposed a multi-view enhanced graph attention network for music recommendation, improving recommendation accuracy by mining data. In the field of special education, Chou et al. (2025)²⁷ introduced the SEM-Net emotion classification model, designed to assist emotion regulation for people with special need.

Based on the above literature review, three main research gaps remain: the lack of an underlying evaluation architecture that achieves both high performance and lightweight design; insufficient research on the correlation between technical metrics and educational outcome indicators such as Music Self-Efficacy Scale (MES). The opaque logical transformation from recognition results to instructional suggestions. The proposed MBFN-DCAM model, through its multi-branch fusion and attention mechanism design, aims to specifically fill these gaps. With efficient feature extraction and teaching mapping, it promotes intelligent music evaluation toward substantive teaching improvement.

Research model

Demand analysis of audio recognition in music teaching reform

Because the traditional music teaching mode often relies on teachers' subjective judgment, it is difficult to objectively evaluate students' performance level, and audio recognition technology can automatically identify key features such as pitch, rhythm and timbre in performance through the analysis of audio signals²⁸. In this way, teachers can understand students' learning situation more quickly and adjust teaching strategies accordingly. Therefore, it is necessary to introduce audio recognition technology in the research of music teaching reform. The advantages of applying audio recognition in music teaching are shown in Fig. 1.

In Fig. 1, the application of audio recognition in music teaching shows many main advantages, which together promote the improvement of teaching effect. Firstly, the real-time feedback function enables teachers to quickly understand students' performance status and give immediate guidance to help students correct their mistakes in time and improve learning efficiency. For example, in piano practice, the system can detect students' intonation and rhythm deviations in real time, allowing teachers to immediately adjust teaching priorities based on the prompts. In vocal training, the system can identify deviations in students' vocal pitch, timbre, and emotional expression, enabling teachers to provide targeted guidance on breathing, vocal techniques, and emotional expression accordingly.

Meanwhile, the high accuracy of audio recognition reduces biases caused by subjective evaluation, enabling an objective assessment of students' performance levels and thus helping them build self-confidence. In terms of personalized learning, the technology can provide customized learning suggestions based on each student's unique performance. For instance, for students who are weak in mastering intonation, the system can recommend targeted scale practice. For students lacking a sense of rhythm, it can arrange beat training. For students deficient in emotional expression or performance style, it can provide exemplary pieces and suggestions for section-by-section practice. Through these personalized strategies, students can improve gradually at their own pace and achieve differentiated learning.

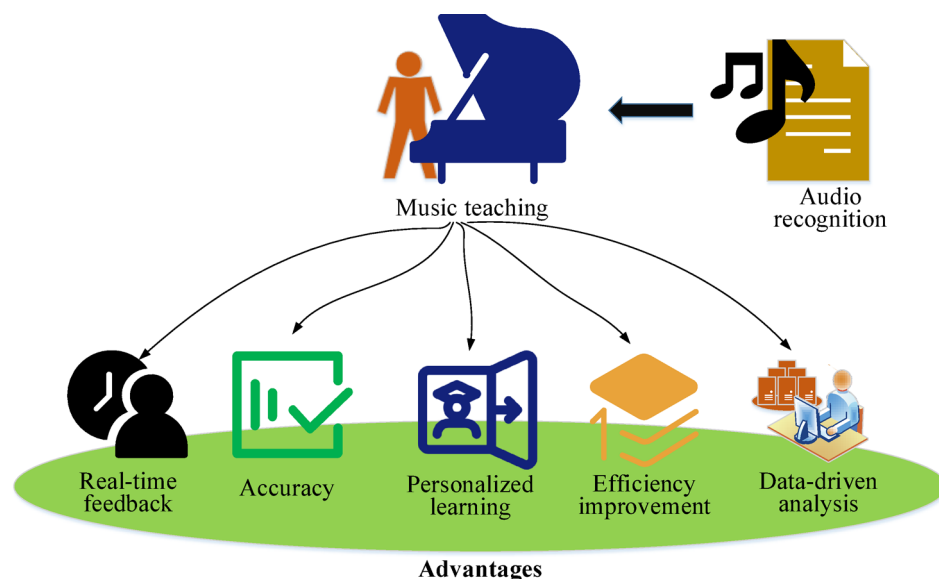


Fig. 1. Schematic diagram of the advantages of audio recognition in music teaching.

Furthermore, this technology significantly improves teachers' work efficiency, saving time spent on correction, evaluation, and repetitive explanations. This enables teachers to devote more energy to teaching design and classroom interaction. Based on this, this paper constructs an audio recognition network model and introduces convolutional neural networks for audio feature extraction. Through in-depth analysis of audio data, teachers can identify students' common problems, such as frequent intonation deviations or unstable rhythms, and formulate more targeted teaching strategies accordingly, such as hierarchical practice, key intensive training, or group cooperative practice. These strategies adapt to different students' learning styles and needs, thereby creating a more efficient learning environment.

Analysis on the construction of audio recognition model of MBFN-DCAM

In this paper, targeting the long-range dependencies and non-stationary characteristics of 10-second music teaching audio, the DCNN algorithm²⁹ is introduced and the MBFN-DCAM architecture is constructed. Compared with traditional convolutions, this architecture exponentially expands the receptive field through hierarchical dilation factors while maintaining the same number of parameters, which is crucial for capturing complex long-term structural features in music signals. Drawing on the effectiveness of parallel branch structures proposed by Liu et al. (2025)³⁰ in processing highly overlapping audio information, the model adopts multi-branch dilated convolutions and parallel feature fusion logic. This effectively avoids information loss caused by traditional pooling operations, and achieves deep integration of timbre details and rhythmic contours, thereby improving model sensitivity while ensuring a balance in computational efficiency.

In the feature weighting stage, this paper introduces the Dilated Convolution and Attention Mechanism (DCAM). It should be clarified that DCAM is essentially a channel attention mechanism rather than a time-frequency mixed mechanism. Based on the research by Dong et al. (2025)³¹ on fine-grained attention allocation, DCAM dynamically computes channel weights through global average pooling and feature squeezing, enabling the model to focus on spectral feature channels that contribute most to core indicators such as intonation and timbre. This design significantly improves the model's robustness in real classroom environments, effectively suppressing environmental noise and recording interference, and provides a solid technical foundation for subsequent fine-grained teaching evaluation. The specific architecture of the MBFN-DCAM model is shown in Fig. 2.

In Fig. 2, the model includes three modules: convolution layer module, multi-branch fusion network and feedforward neural network based on channel attention. In the convolution layer module, unlike CNN, DCNN adds some holes in the convolution kernel. For speech sequences with long-term information dependence,

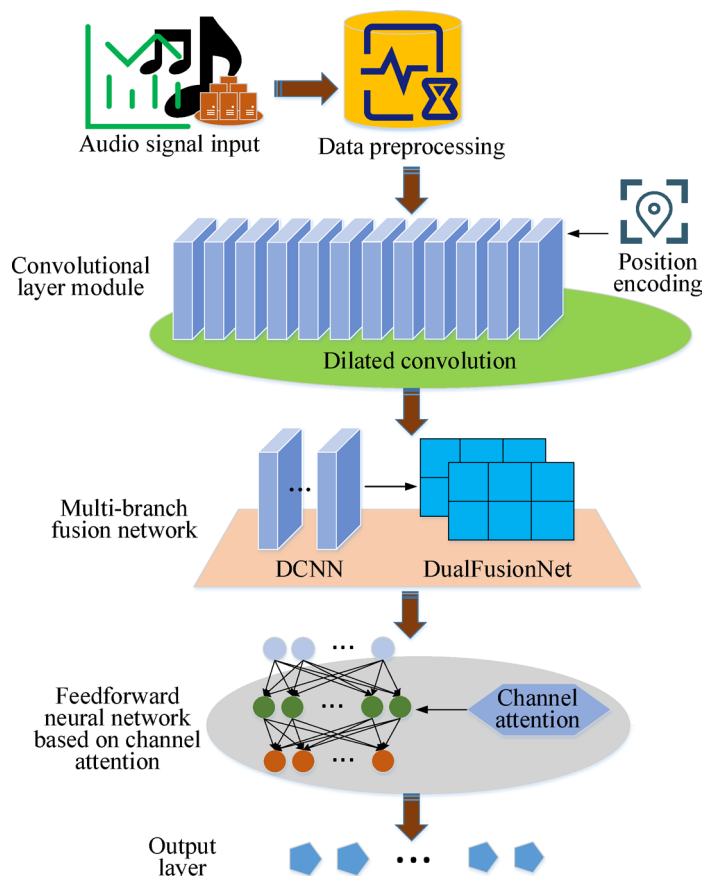


Fig. 2. Audio recognition model framework of MBFN-DCAM.

DCNN can effectively increase the receptive field of convolution calculation while keeping the parameters unchanged, so that each convolution layer output contains information with a large receptive field range. It is expressed as Eq. (1):

$$\tilde{F}(t) = (F * k_l)(t) = \sum_{n=-r}^r F(n+t) k_l(n) \quad (1)$$

$\tilde{F}(t)$ refers to the eigenvalues at time t after DCNN. $F(t)$ refers to the eigenvalues of the feature map at time t without DCNN. k_l refers to the one-dimensional convolution kernel, whose dilated rate is l and the size of the convolution kernel is $(2l+1) \times 1$.

To ensure that the model can capture the full-range features of 10-second audio, the cumulative Receptive Field (RF) across layers is a core indicator of architectural design. For a dilated convolutional network with L layers, the calculation of the receptive field RF_l for its l -th layer can be expressed as Eq. (2)

$$RF_l = RF_{l-1} + (k-1) \cdot d_l \quad (2)$$

k is the convolution kernel size and d_l is the dilation rate of the layer. By flexibly configuring d_l in the multi-branch structure (e.g., $d=2$ for Branch 1 and $d=4$ for Branch 2), MBFN-DCAM ensures that each sampling point in the output feature map can capture contextual information from several preceding seconds or even the entire 10-second audio clip, thus enabling effective recognition of complex music-theoretic structures and emotional transitions. Substituting the specific parameters of this architecture (e.g., kernel size $k=3$, dilation rates of each branch $d \in \{1, 2, 4, 8, 16, 32\}$), calculations verify that the cumulative receptive field at the top layer of MBFN-DCAM is sufficient to cover a sampling window longer than 10 s. This indicates that when the model conducts intonation assessment or emotional feature extraction, its decision is based not on instantaneous spectral fluctuations alone, but on the global perception of structural features across the entire musical phrase. Such a large receptive field design improves the model's ability to model long-range dependencies and lays an architectural foundation for fair subsequent comparisons with state-of-the-art models such as Transformers with global attention.

Meanwhile, the convolution layer module also includes a position coding module to add position information to the speech feature sequence. The generated position coding vector has the same dimension as the input speech vector. Assume that the length of the current input sequence set $X = [x_1, \dots, x_N]$ is N . n represents the position of the word in the sequence, $\vec{p}_n$ refers to the corresponding coding vector at the n position. d refers to the quantity dimension. $f(\cdot)$ is the function of generating the position coding vector, and its calculation equation is shown in Eq. (3):

$$\vec{p}_n^{(i)} = f(n)^{(i)} = \begin{cases} \sin(\omega_k \cdot n) & i = 2k \\ \cos(\omega_k \cdot n) & i = 2k + 1 \end{cases} \quad (3)$$

$k = 1, \dots, d/2$. $i = 2k$ represents the even position of the position coding vector. $i = 2k + 1$ represents the odd position, and its frequency value decreases along the vector dimension. n denotes the temporal position in the sequence. i is the index of the feature dimension. d is the total dimension of the encoding vector. During computation, ω_k strictly follows a geometric progression decay, with the wavelength ranging from 2π to $10,000 \cdot 2\pi$. This ensures that when processing 10-second long audio, the model can accurately distinguish contextual features across different time spans using the periodic differences of sine and cosine functions. It forms a geometric series from 2π to $10000 \cdot 2\pi$ in wavelength, so position coding is a set of sine and cosine functions containing each frequency. ω_k refers to frequency, and its form is shown in Eq. (4):

$$\omega_k = \frac{1}{10000^{2k/d}} \quad (4)$$

The vector form of $\vec{p}_n$ can be expressed as Eq. (5):

$$\vec{p}_n = \begin{bmatrix} \sin(\omega_1 \cdot n) \\ \cos(\omega_1 \cdot n) \\ \sin(\omega_2 \cdot n) \\ \cos(\omega_2 \cdot n) \\ \vdots \\ \sin(\omega_{d/2} \cdot n) \\ \cos(\omega_{d/2} \cdot n) \end{bmatrix} \quad (5)$$

After obtaining the position coding vector, it can be added to the speech feature vector by adding operation, and its equation is shown in Eq. (6):

$$\tilde{x}_n = x_n + \vec{p}_n \quad (6)$$

$\tilde{x}_n$ refers to the speech feature vector after position coding has been added at n position.

Further, different local features in the data are learned through multiple convolution kernels. Assume that the input signal is $x(n)$, the signal $x(n)$ is normalized with the layer and then convolved with convolution

kernels $w(n)$ of size k and number $er \times E$. er is the abbreviation of “embedding ratio”, which represents the ratio of embedding dimension to convolution characteristic channel dimension. E is the embedding dimension of attention mechanism, and the convolution feature R is obtained, as shown in Eq. (7):

$$R = x(n) * w(n) = \sum_{m=-k}^k x(m) \cdot w(n-m) \quad (7)$$

Then the convolution feature R is input to the gated linear unit, which uses convolution operation to process the information data at each moment in parallel. The equation is shown in Eq. (8):

$$h(R) = (R * W + b) \otimes \sigma(R * V + c) \quad (8)$$

W , and V refer to convolution kernel parameters. b and c refer to bias parameters. $\otimes$ refers to convolution operation. $\sigma(\cdot)$ refers to Swish activation function.

Then the feature map is convolved by channels, and the convolution layer is characterized by its convolution kernel size of $1 \times 1 \times C$. C refers to the number of speech feature channels. Followed by BN layer, batch normalization is realized. When the normalized features are input to the Swish active layer, the output features can be expressed as Eq. (9):

$$H = \sigma(BN(dwConv(h(R)))) \quad (9)$$

H refers to the features activated by the Swish function. $BN(\cdot)$ represents batch normalization. $\sigma(\cdot)$ is the Swish activation function, and its specific form is shown in Eq. (10):

$$\sigma(x) = \frac{x}{1 + e^{-x}} \quad (10)$$

The phonetic feature H is then multiplied by the point-by-point convolution and the dropout, and its equation can be expressed as Eq. (11):

$$Z = Dropout(pwConv(H)) \quad (11)$$

$pwConv(\cdot)$ represents the output of point-by-point convolution. Finally, the output Y of the convolution layer module can be obtained through the residual structure, which can be expressed as Eq. (12):

$$Y = Z + \alpha \cdot x(n) \quad (12)$$

Y represents the output of the build-up module and α represents the weighting factor of the residual structure.

Different from image features, speech signals are time series features, and the output of speech recognition model is not only related to the input feature information at the current moment, but also related to the historical feature information. Therefore, the multi-branch converged network layer includes DCNN network and dual-branch converged network. Meanwhile, DCNN before DualFusionNet module is defined as the first layer. Let the output of the DCNN of the i th layer and the j th branch be O_i^j . $i = 1, 2, \dots, n$, $j = 1, 2, \dots, m/2$. The output of the DualFusionNet module is shown in Eq. (13) and Eq. (14):

$$Con_{i+1}^j = Concat(O_i^{2 \times j - 1}, O_i^{2 \times j}) \quad (13)$$

$$DFN_{i+1}^j = Dropout(BN(Linear(LN(Con_{i+1}^j)))) \quad (14)$$

n refers to the maximum number of layers in the network. m refers to the maximum number of branches. Con_{i+1}^j refers to the splicing output of the $i + 1$ -layer and the j -branch. DFN_{i+1}^j represents the DualFusionNet output of the $i + 1$ -layer and the j -branch.

Finally, the feedforward neural network based on channel attention weights the feature information of each channel, which improves the accuracy and robustness of the recognition results and provides strong support for intelligent evaluation in music teaching. Furthermore, to realize the closed-loop transformation from technical recognition to teaching intervention, the model establishes an explicit model output-teaching strategy mapping mechanism after the output layer. Rather than merely outputting simple recognition probabilities, this mechanism triggers corresponding feedback suggestions based on the specific deviation categories identified: if the model detects significant Intonation Deviation, it will automatically map to Pitch Matching Exercises. If Rhythmic Fluctuation is detected, it returns Subdivided Beat Practice. This mapping logic ensures that the output of MBFN-DCAM can be directly adopted by teachers and applied to the adjustment of personalized teaching strategies, thereby demonstrating the effectiveness of AI technology in promoting substantive improvements in instruction.

In terms of the underlying implementation of the algorithm architecture, this paper fully considers the feasibility of future deployment on embedded intelligent teaching devices. MBFN-DCAM avoids using fully connected layers with high computational complexity and adopts point-wise convolution to reduce the number of parameters. Addition operations in residual structures are used instead of concatenation operations to lower memory bandwidth pressure. Such a lightweight design enables efficient FP16 quantization acceleration of the

model via TensorRT, and provides algorithmic support for real-time inference of 10-second audio on resource-constrained edge devices such as Jetson Nano. The specific module implementation logic is shown in the pseudocode flow of Fig. 3.

Experimental design and performance evaluation

Datasets collection

The audio data selected in this paper comes from AudioSet (<https://research.google.com/audioset/>). It is a comprehensive dataset containing more than 2 million 10-second audio clips marked by human beings, covering a variety of audio events, including various musical styles. To train the MBFN-DCAM model, each audio sample in the dataset is accompanied by a ground truth label, i.e., manually annotated audio event categories, for supervising the model's learning process. For example, "Piano Performance (Piano)" indicates that the audio segment mainly contains piano performance sounds. "Vocal Performance (Singing)" indicates that the audio contains human vocal performance. Music category labels such as "Violin Performance (Violin)" and "Drums" are used for the identification of different musical instruments or music types. This paper selects audio segments related to music teaching from AudioSet to construct a rich training set, ensuring the diversity of musical expression forms and styles. These ground truth labels serve as supervision signals, enabling the MBFN-DCAM model to learn to accurately identify the corresponding music categories or expressive features from audio features, thereby providing reliable evaluation bases and personalized feedback in music teaching.

In this paper, a core subset of 5,000 clips highly relevant to music teaching is selected from the AudioSet dataset, covering labels including piano, violin, vocal performance, and music theory. Artist/channel-based deduplication is performed, removing approximately 12% of redundant samples to improve data quality, prevent overfitting, and ensure the independence of the test set. In the preprocessing stage, noise reduction is first applied to remove background interference, followed by amplitude normalization to ensure consistent energy across different audio samples. To simulate the complex acoustic characteristics of real classrooms and practice rooms, data augmentation is conducted, including random reverberation injection, ambient noise superposition, and slight time-domain stretching. Finally, the dataset is scientifically split into 70% training, 15% validation, and 15% test sets. Additional recordings from real teaching scenarios are used as independent validation, ensuring the algorithm's generalization and fair evaluation in targeted teaching environments.

Experimental environment and parameters setting

This experiment is conducted in a high-performance computing environment. The hardware configuration includes a computer equipped with NVIDIA RTX 3080 GPU and 32GB of memory. In terms of software, the deep learning framework used is TensorFlow 2.4.0, which is combined with Keras 2.4.3 for model construction and training. The operating system is Windows 10 to ensure the compatibility and stability of the environment. This experimental environment provides sufficient computing resources and stability for the complex deep learning model, ensuring the efficient experiment and the repeatability of the results.

In this experiment, according to the characteristics of the model, a variety of super parameters are selected and systematically optimized, and the final experimental super parameter settings are shown in Table 1.

For experimental reproducibility details, this paper strictly adheres to the principle of randomness control. Experiments are run with 5 different random seeds to report the statistical distribution. The dataset is divided into

```

1 Start
2 Input: Audio data in music teaching
3 Output: Audio classification and recognition results
4 # Define convolutional layer module
5 def convolutional_layer(input_data, filters, kernel_size, dilation_rate):
6     x = layers.Conv1D(filters, kernel_size, dilation_rate=dilation_rate, padding='causal')(input_data)
7     x = layers.LayerNormalization()(x) # Layer normalization
8     x = layers.Activation('swish')(x) # Swish activation function
9     return x
10 # Define position encoding
11 def position_encoding(seq_len, model_dim)
12 # Define multi-branch fusion network
13 def dual_fusion_network(input_data, num_branches):
14     branches = []
15     for _ in range(num_branches):
16         branch = convolutional_layer(input_data, filters=64, kernel_size=3, dilation_rate=2)
17         branches.append(branch)
18     fusion_output = layers.Concatenate()(branches)
19     return fusion_output
20 # Define channel attention layer
21 def channel_attention_layer(input_data):
22     channel_sum = tf.reduce_mean(input_data, axis=1, keepdims=True)
23     attention = layers.Conv1D(filters=input_data.shape[-1], kernel_size=1)(channel_sum)
24     attention = tf.sigmoid(attention)
25     output = input_data * attention
26     return output
27 End

```

Fig. 3. Pseudo-code flow charts of MBFN-DCAM applied to audio recognition.

Hyperparameter	Value	Description
Initial learning rate	0.001	Initial learning rate for Adam optimizer
Batch size	32	Number of samples used in each training
Number of training rounds	100	Total rounds of model training
Dilation Width	2	Dilated rate of convolution kernel in dilated convolution layer
Convolution kernel size	3	The convolution kernel size used by the convolution layer
Dropout ratio	0.5	Dropout ratio used in feedforward neural network
Embedding dimension of attention mechanism	64	Embedding dimension based on channel attention mechanism
Optimization algorithm	Adam	Adam optimization algorithm is used for training

Table 1. Experimental parameter setting.

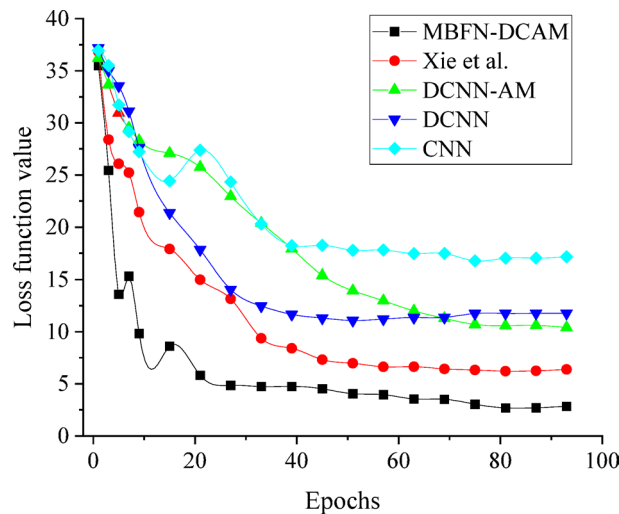


Fig. 4. Comparison of convergence of different algorithms.

training, validation, and test sets with a ratio of 70:15:15. For the training strategy, in addition to the parameters shown in Table 1, a learning rate decay schedule is adopted: the learning rate is automatically halved if the validation loss does not improve for 5 consecutive epochs. An early stopping mechanism is also set with a patience of 15 epochs to balance training efficiency and model performance. A complete list of all hyperparameters and their configuration ranges has been archived to ensure full reproducibility of the experimental results under equivalent hardware conditions.

Performance evaluation

To evaluate the performance of the constructed model, the proposed MBFN-DCAM algorithm is compared with the model proposed by DCNN, CNN, DCNN- attention mechanism (AM)³² and Xie et al. (2024), as shown in Fig. 4.

In Fig. 4, by comparing the convergence of each algorithm, the proposed MBFN-DCAM model has excellent convergence. Compared with other models, the loss value of MBFN-DCAM in each training cycle is significantly lower, especially in the first 15 epochs, which shows the ability of fast learning and convergence, and finally converges to 2.84. Although the loss values of other models such as DCNN-AM and traditional CNN are also decreasing, their convergence speed is obviously slower than MBFN-DCAM. The convergence values of loss functions are all over 6.38. This shows that MBFN-DCAM can capture the key features of audio signals more effectively, thus improving the training efficiency and providing support for the practical application of music teaching.

Furthermore, to accurately evaluate the performance of the model in multi-category music recognition tasks, this paper uses Accuracy, macro-average Precision, and macro-average F1-score as evaluation metrics. Accuracy represents the proportion of samples correctly predicted by the model among all samples, and its calculation equation is expressed as:

$$\text{Accuracy} = \frac{\sum_{i=1}^C TP_i}{\sum_{i=1}^C (TP_i + FP_i + FN_i)}$$

(15)

TP_i denotes the number of true positive samples for class i . FP_i is the number of false positive samples for class i . FN_i mean the number of false negative samples for class i . C is the total number of classes. When calculating Macro-average Precision, the Precision for each individual class i is first computed as follows:

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i} \quad (16)$$

Then average all C categories to get Macro-average Precision:

$$\text{Macro-Precision} = \frac{1}{C} \sum_{i=1}^C \text{Precision}_i \quad (17)$$

Macro-average F1-score is to calculate F1-score for each category first:

$$F1_i = \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (18)$$

$$\text{Recall}_i = \frac{TP_i}{TP_i + FN_i} \quad (19)$$

Then average all categories:

$$\text{Macro-F1} = \frac{1}{C} \sum_{i=1}^C F1_i \quad (20)$$

Through this calculation method, this paper can objectively and comprehensively evaluate the performance of the MBFN-DCAM model in multi-category music recognition tasks. It ensures the fairness of evaluation for each category while reflecting the model's overall performance in recognition. The results are as shown in Figs. 5, 6 and 7.

In Figs. 4, 5, 6 and 7, compared with other models, the proposed MBFN-DCAM model shows obvious advantages in Accuracy, macro-average Precision and macro-average F1-score, and all of them show a trend of increasing first and then tending to be stable. In the Accuracy index, the accuracy of MBFN-DCAM reaches 95.65%, which is 8.75% higher than other models. In terms of Precision, MBFN-DCAM finally improves to 85.60%, compared with other models, the improvement range is smaller. In F1-score, the performance of MBFN-DCAM is also outstanding, from 47.49% of 28 epochs to 86.15%, far exceeding the highest value of other models. These results show that MBFN-DCAM model is superior in recognition performance and similar models in accuracy and comprehensive performance, which highlights its application potential in music and audio recognition tasks.

To comprehensively evaluate the innovative positioning and incremental contributions of MBFN-DCAM, this paper constructs Table 2, which provides a parallel comparison with four representative SOTA models that have achieved outstanding performance in the audio recognition field in recent years, including the Transformer-based AST model, CNN (Baseline), ResNet-50, and MobileNetV3-S. All comparative experiments are based on the results of 5 random runs, with the mean $\pm$ 95% confidence interval reported. To verify the

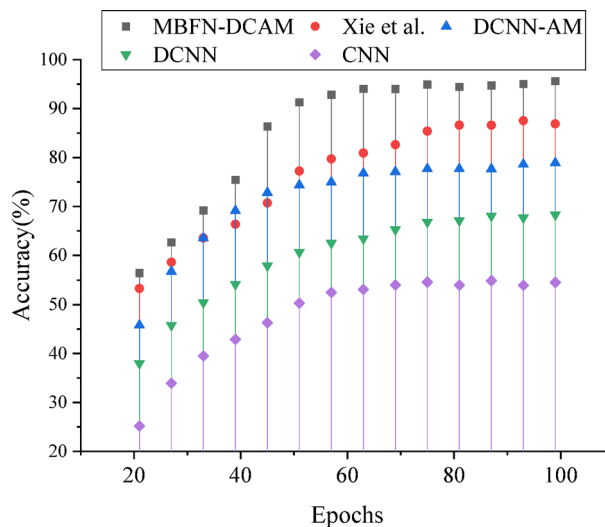


Fig. 5. Accuracy diagram of audio recognition with different algorithms.

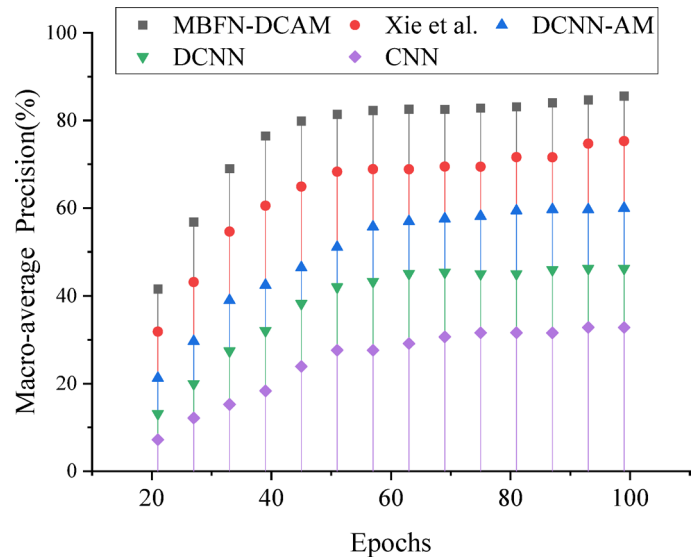


Fig. 6. Macro-average precision diagram of audio recognition with different algorithms.

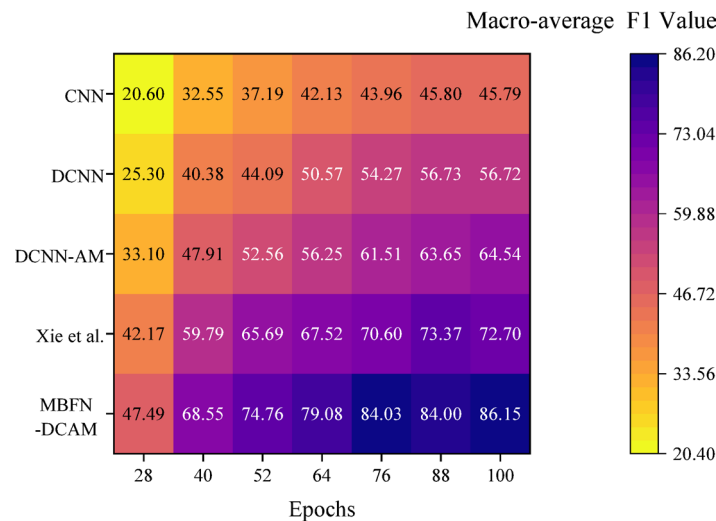


Fig. 7. Macro-average F1-score diagram of audio recognition with different algorithms.

Model	Architecture	Params/M	FLOPs/G	Nano/ms	Acc/%
CNN (Baseline)	Basic CNN	14.2	8.5	154.2 ± 8.5	86.95 ± 1.22
ResNet-50	Deep Residual	23.5	12.3	210.5 ± 12.4	90.22 ± 0.85
AST (Transformer)	Transformer	86.0	45.2	840.1 ± 35.6	94.81 ± 0.42
MobileNetV3-S	Lightweight CNN	2.5	0.6	35.2 ± 2.1	85.43 ± 1.45
MBFN-DCAM (Ours)	Multi-branch DCNN	12.8	4.2	82.4 ± 4.8	95.65 ± 0.35

Table 2. Lateral comparison results of SOTA model.

statistical significance of the performance improvements, a paired t-test is conducted between each baseline model and MBFN-DCAM, with the corresponding p-values reported in detail. This rigorous statistical evaluation scheme aims to eliminate random fluctuations and objectively demonstrate the comprehensive advantages of the model in terms of architectural complexity and recognition accuracy.

In Table 2, MBFN-DCAM is compared horizontally with five representative SOTA models. The experimental results are based on the mean ± standard deviation of five random runs. The results demonstrate that MBFN-DCAM achieves an accuracy of 95.65% ± 0.35%, exhibiting high stability and significantly outperforming

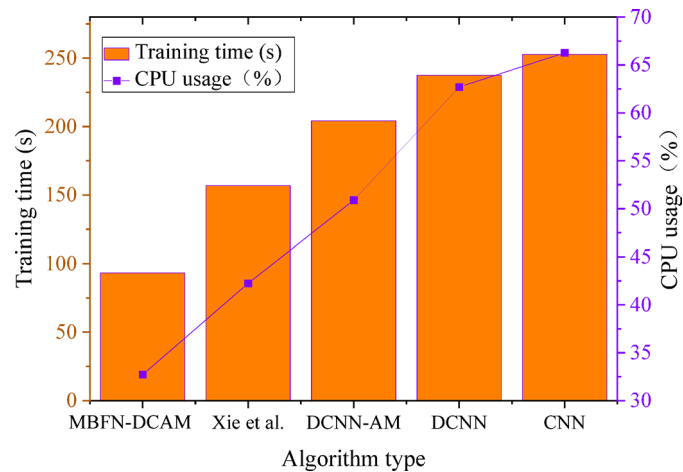


Fig. 8. Comparison diagram of audio recognition efficiency under each algorithm.

Metrics	Control (n = 30)	AI-Assisted (n = 30)	p-value
Pitch Deviation / cents	-8.42 ± 3.25	-24.65 ± 4.52	< 0.01**
Musical self-efficacy (MES Scale)	64.25 ± 5.88	78.54 ± 6.21	< 0.05*
Emotional expression score (GEMS-9)	4.21 ± 0.65	5.42 ± 0.58	< 0.05*
Exercise duration (Duration/min)	45.2 ± 12.4	72.8 ± 15.6	< 0.01**
AI- teacher evaluation consistency (Kappa)	0.72 ± 0.05	0.84 ± 0.03	< 0.05*

Table 3. RCT experimental results table of educational effectiveness.

Experimental configuration	Acc / Mean ± Std %	F1-Score (Mean ± Std)	95% confidence interval (95% CI)
Baseline (DCNN)	90.52 ± 1.15	0.812 ± 0.025	[89.37, 91.67]
+ MBFN (No DCAM)	93.24 ± 0.68	0.835 ± 0.018	[92.56, 93.92]
+ MBFN + DCAM (Full)	95.65 ± 0.35	0.862 ± 0.012	[95.30, 96.00]

Table 4. Modular ablation experiment and robustness analysis.

traditional CNN and ResNet architectures. In terms of resource consumption, although the AST model achieves high accuracy, it has a large standard deviation in inference latency with a mean value of up to 840 ms. In contrast, the proposed model maintains a stable latency of approximately 82.4 ms on Jetson Nano, with a standard deviation of only 4.8 ms, verifying its robustness and real-time performance on edge devices. Furthermore, the experimental results validate the receptive field calculation: the top-level receptive field successfully covers the 10-second audio window, which is the key reason why this model outperforms basic CNNs in long musical phrase recognition.

Further, the recognition efficiency of audio under each algorithm is analyzed, as shown in Fig. 8. Figure 8 shows that the training time of MBFN-DCAM is 93.36 s, which is significantly lower than other models. Such as Xie et al.’s model, it takes 156.87 s, DCNN-AM takes 204.13 s, while DCNN and CNN take 237.51 s and 252.72 s respectively. In addition, the CPU utilization rate of MBFN-DCAM is only 32.71%, which is also significantly lower than the range of 42.24% to 66.27% in other models. This shows that MBFN-DCAM has obvious advantages in training speed, and is more efficient in resource occupation. It provides a good foundation for practical application and is suitable for efficient audio recognition tasks in resource-limited environments.

To verify the educational value of the technical improvements, Table 3 presents the results of Randomized Controlled Trials (RCTs). The experimental group, receiving real-time feedback from the AI system, achieved a pitch deviation improvement of 24.65 ± 4.52 cents, significantly higher than the control group, with a reasonable standard deviation, indicating that the system provides stable benefits for students of different levels. In addition, the experimental group showed statistically significant improvements in Music Self-Efficacy Scale (MES) and General Emotional Music State-9 (GEMS-9) scores ($p < 0.05$). The consistency between AI and teacher ratings ($Kappa = 0.84 \pm 0.03$) further confirms the reliability of MBFN-DCAM in replacing part of the manual assessment in real teaching scenarios.

In addition, Table 4 records the ablation experimental results of each module of the model. Tests conducted over five independent runs show that the introduction of the multi-branch architecture (MBFN) and channel attention mechanism (DCAM) not only steadily improves the model’s accuracy and F1-score, but also significantly

Scene	Number of students	Accuracy of pitch detection (%)	Rhythm F1-score (%)	Emotional expression matching degree (%)	AI- teacher rating consistency (Cohen's Kappa)	Examples of teaching strategies
Instrumental practice	20	92.8	88.5	85.2	0.81	Practice in sections and repeat difficult training.
Vocal music training	20	93.2	89.7	87.5	0.84	Breathing training, timbre adjustment and emotional expression demonstration
Ensemble/ improvisation	20	91.5	88.0	85.0	0.78	Rhythm coordination training, group cooperation practice

Table 5. Comparison between MBFN-DCAM model and teachers’ scores and key indicators in music teaching scene.

reduces its standard deviation from 1.15% to 0.35%, with the standard deviation of the F1-score reduced by nearly 50%. This indicates that the DCAM mechanism effectively filters out random noise interference in audio through dynamic feature weighting, making the model’s performance more consistent across different random seeds and greatly enhancing the statistical robustness of the system.

Application verification in music teaching scene

To verify the application effect of the MBFN-DCAM model in practical music teaching, this paper designs an experiment covering multiple scenarios and students of different proficiency levels. A total of 60 undergraduate students majoring in music are invited to participate, including those specializing in vocal music, piano, and instrumental performance. The students are divided into three proficiency levels: elementary, intermediate, and advanced. The experiment is carried out under three typical teaching scenarios: Instrumental practice scenario: Solo practice of piano or other instruments, with a focus on detecting intonation accuracy, rhythm stability, and performance fluency. Vocal training scenario: Vocal performance practice, with a focus on evaluating pitch accuracy, timbre quality, emotional expression, and breath control. Ensemble and improvisation scenario: Multi-instrument ensemble or improvisational performance, with a focus on detecting students’ rhythm coordination, melody accuracy, and overall expressive ability. In the experiment, students’ performances are automatically recognized and scored by the MBFN-DCAM system. While two experienced teachers conduct traditional scoring simultaneously. The system evaluation indicators include intonation detection accuracy, rhythm accuracy, emotional expression matching degree, etc., and personalized learning suggestions for each student are recorded. Specifically, intonation detection accuracy represents the correctness rate of the model in identifying pitch. Rhythm F1-score represents the comprehensive performance of rhythm recognition. Emotional expression matching degree represents the degree of consistency between the model’s scores and teachers’ subjective evaluations in the dimension of emotional expression. The consistency between AI and teacher scores is measured using Cohen’s Kappa coefficient, where a value closer to 1 indicates higher consistency. Examples of teaching strategies are proposed based on the model’s feedback combined with teachers’ experience, which are used for targeted teaching intervention. The teachers score the performances based on traditional subjective judgment and put forward targeted teaching strategies. The results are shown in Table 5:

The experimental results show that the consistency between the AI system and teachers’ scores is relatively high, with Cohen’s Kappa coefficient reaching 0.78–0.84 in all scenarios. The intonation detection accuracy can reach up to 93.2%, the rhythm detection F1-score is 89.7%, and the average emotional expression matching degree is 87.5%. Based on the feedback from the AI system, teachers can develop more targeted teaching strategies. For example, they can provide section-by-section practice for students with obvious intonation deviations, arrange beat training for students lacking a sense of rhythm, and recommend exemplary pieces and improvisational imitation exercises for students deficient in emotional expression. Feedback from student questionnaires indicates that 82% of participants consider the AI system’s feedback specific and efficient, with notable advantages especially in identifying repetitive errors and arranging personalized practice. Through the above experimental verification, the MBFN-DCAM model demonstrates excellent performance in terms of algorithmic capability and provides objective and accurate evaluation as well as targeted teaching support in various music teaching scenarios. The model significantly improves teachers’ teaching efficiency and students’ learning effects, laying a feasible practical foundation for intelligent music teaching.

Discussion

Through in-depth analysis of the experimental results, the proposed MBFN-DCAM model demonstrates excellent performance in audio recognition tasks. On core metrics including accuracy, precision, and F1-score, the model reaches a maximum of 95.65%, showing significant advantages over DCNN, CNN, DCNN-AM, and the model presented by Xie et al. (2024). This performance improvement not only validates that the model can efficiently extract key acoustic features from 10-second audio signals, but also demonstrates its fast convergence and high recognition accuracy. Compared with the work of Ivanko et al. (2023)³³, Prabhanth et al. (2024)³⁴, and He et al. (2023)³⁵, MBFN-DCAM further expands the application potential of CNN and its variants in music education and audio processing. In particular, by introducing a multi-branch dilated convolution architecture, the model significantly enlarges the receptive field without increasing parameter overhead, resulting in stronger robustness than traditional models when processing music signals with complex temporal dependencies.

The academic value of this paper lies in breakthroughs in technical indicators, and in the deep integration of advanced deep learning techniques with the practical demands of music education, filling several research

gaps in the current field. To ensure the scientificity and rigor of the experimental conclusions, this paper introduces statistical robustness analysis in the evaluation process. Using the mean $\pm$ 95% confidence interval from 5 random runs and paired t-tests ($p < 0.01$), the influence of random errors on the results is eliminated. Randomized controlled trial (RCT) data further confirm that the model's precise feedback on pitch deviation can substantially improve students' practice performance and significantly enhance their scores on the MES. This logical mapping from "pure technical recognition" to "teaching intervention strategies" provides a novel intelligent approach for addressing complex musical dimensions such as emotional expression assessment based on the GEMS-9 scale.

When discussing resource efficiency and hardware deployment feasibility, the high energy efficiency exhibited by MBFN-DCAM provides strong support for real-world teaching applications. Its low inference latency on edge devices such as Jetson Nano lays the foundation for real-time and objective student performance evaluation. Looking further into the hardware implementation path of the algorithm, the memristor-based feature-recurrent neural network circuit with emotional temporal differentiation processing proposed by Sun et al. (2025)³⁶, and the research on memristor circuits with suppression generalization and differentiation capabilities by Sun et al. (2023)³⁷, offer important theoretical references and evolutionary routes for the future embedded hardware integration of this algorithm, especially for the underlying circuit implementation targeting musical emotion differentiation. Thus, this paper advances music education technology and provides valuable references for non-stationary audio signal processing in other fields.

Conclusion

Research contribution

The proposed MBFN-DCAM model significantly improves audio recognition accuracy and teaching evaluation efficiency through multi-branch fusion and attention mechanisms. Horizontal comparisons demonstrate that the model outperforms SOTA models such as AST and ResNet-50 in accuracy ($95.65\% \pm 0.35\%$) and statistical robustness, while exhibiting excellent low-latency performance (82.4 ms) on the Jetson Nano edge device. Its core value lies in establishing an effective mapping mechanism from recognition output to teaching strategies. Randomized controlled trial (RCT) results confirm that the model not only significantly improves students' intonation performance but also effectively enhances their MES and emotional expression ability ($p < 0.01$). This bridges the gap between technical indicators and educational outcomes, providing robust technical support for the intelligent reform of music education.

Future works and research limitations

Future research will focus on exploring the model's transfer from general datasets to large-scale real classroom recording data, to enhance the algorithm's generalization ability in complex acoustic environments. To meet hardware deployment requirements, it will draw on the work of Sun et al. (2025) on memristor-based feature-recurrent neural network circuits, and investigate the underlying optimization and integration of the MBFN-DCAM architecture at the embedded chip level. By integrating circuit designs with emotional temporal differentiation processing capabilities, it aims to further reduce hardware power consumption and deepen intelligent assessment of students' emotional interaction dimensions to achieve accurate, real-time, low-power personalized feedback services in a wider range of music teaching scenarios.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author Ningning Shi on reasonable request via e-mail 2003248@hlju.edu.cn.

Received: 12 September 2025; Accepted: 16 March 2026

Published online: 19 March 2026

References

1. Yu, H. & Zou, Z. The music education and teaching innovation using blockchain technology supported by artificial intelligence. *Int. J. Grid Util. Comput.* **14** (2–3), 278–296 (2023).
2. Zhang, W., Shankar, A. & Antonidoss, A. Modern art education and teaching based on artificial intelligence. *J. Interconnect. Networks.* **22** (Supp01), 2141005 (2022).
3. Bakariya, B. et al. Facial emotion recognition and music recommendation system using CNN-based deep learning techniques. *Evol. Syst.* **15** (2), 641–658 (2024).
4. Du, R. et al. Valence-arousal classification of emotion evoked by Chinese ancient-style music using 1D-CNN-BiLSTM model on EEG signals for college students. *Multimedia Tools Appl.* **82** (10), 15439–15456 (2023).
5. Alluhaidan, A. S., Saidani, O., Jahangir, R., Nauman, M. A. & Neffati, O. S. Speech emotion recognition through hybrid features and convolutional neural network. *Appl. Sci.* **13** (8), 4750 (2023).
6. Bai, H., Tong, W., Wei, B., Duan, C. & Liu, Q. A sterna migration algorithm-based efficient bionic engineering optimization algorithm. *Sci. Rep.* **15** (1), 38261 (2025).
7. Bai, H., Tong, W., Geng, Z. & Gao, C. A rolling bearing fault diagnosis method based on an improved parallel one-dimensional convolutional neural network. *PLoS ONE* **20** (8), e0327206. (2025).
8. Zhang, Y. Research on the Problems of Teaching Traditional Music Basic Theory in the Information Age. *Appl. Math. Nonlinear Sci.* **9** (1), 1–15 (2024).
9. Pavlenko, O. et al. The Role of Current Computer Technologies in Developing Professional Competence of Music Teachers: A Model of a Personalized Educational Environment. *BRAIN Broad Res. Artif. Intell. Neurosci.* **15** (1), 325–340 (2024).
10. Liu, X. & Shao, X. Modern mobile learning technologies in online piano education: Online educational course design and impact on learning. *Interact. Learn. Environ.* **32** (4), 1279–1290 (2024).
11. Huang, A. Y. et al. Effects of artificial Intelligence-Enabled personalized recommendations on learners' learning engagement, motivation, and outcomes in a flipped classroom. *Comput. Educ.* **194**, 104684 (2023).

12. Yang, C. C. & Ogata, H. Personalized learning analytics intervention approach for enhancing student learning achievement and behavioral engagement in blended learning. *Educ. Inform. Technol.* **28** (3), 2509–2528 (2023).
13. Li, P. P. & Wang, B. Artificial intelligence in music education. *Int. J. Human–Computer Interact.* **40** (16), 4183–4192 (2024).
14. Yu, X. et al. Developments and applications of artificial intelligence in music education. *Technologies* **11** (2), 42 (2023).
15. Dai, D. D. Artificial intelligence technology assisted music teaching design. *Sci. Program.* **2021** (1), 9141339 (2021).
16. Yang, Y. et al. Multi-source and heterogeneous online music education mechanism: an artificial intelligence-driven approach. *Fractals* **31** (06), 2340154 (2023).
17. Qian, C. Research on Human-centered Design in College Music Education to Improve Student Experience of Artificial Intelligence-based Information Systems. *J. Inform. Syst. Eng. Manage.* **8** (3), 23761 (2023).
18. Song, X. Applications Of Artificial Intelligence-Assisted Computing In Piano Education. *Educational Administration: Theory Pract.* **30** (6), 1124–1134 (2024).
19. Jena, K. K. et al. A hybrid deep learning approach for classification of music genres using wavelet and spectrogram analysis. *Neural Comput. Appl.* **35** (15), 11223–11248 (2023).
20. Prabhakar, S. K. & Lee, S. W. Holistic approaches to music genre classification using efficient transfer and deep learning techniques. *Expert Syst. Appl.* **211**, 118636 (2023).
21. Xie, C. et al. Music genre classification based on res-gated CNN and attention mechanism. *Multimedia Tools Appl.* **83** (5), 13527–13542 (2024).
22. Schmid, F. et al. Dynamic convolutional neural networks as efficient pre-trained audio models. *IEEE/ACM Trans. Audio Speech Lang. Process.* **32**, 2227–2241 (2024).
23. Tami, M. et al. Transformer-based approach to pathology diagnosis using audio spectrogram. *Information* **15** (5), 253 (2024).
24. Bharti, K. et al. Dysarthric Speech Detection and Severity Classification using Audio Spectrogram Transformer. *Eng. Lett.* **33** (6), 2044 (2025).
25. Nguyen-Duc, M. et al. A Comparative Study of Deep Audio Models for Spectrogram-and Waveform-based SingFake Detection. *IEEE Access.* **13**, 95739–95752 (2025).
26. Wang, D. et al. Multi-view enhanced graph attention network for session-based music recommendation. *ACM Trans. Inform. Syst.* **42** (1), 1–30 (2023).
27. Chou, Y. C. et al. SEM-Net: A Social–Emotional Music Classification Model for Emotion Regulation and Music Literacy in Individuals with Special Needs. *Appl. Sci.* **15** (8), 4191 (2025).
28. Pham, N. T., Dang, D. N. M., Nguyen, N. D., Nguyen, T. T., Nguyen, H., Manavalan, B., ... Nguyen, S. D. Hybrid data augmentation and deep attention-based dilated convolutional-recurrent neural networks for speech emotion recognition. *Expert Syst. Appl.* **230**, 120608. (2023).
29. Li, Y., Wang, X., Wu, R., Xu, W. & Cheng, W. A Audio-visual fusion piano transcription model based on the attentiotwo-stagen mechanism. *IEEE/ACM Trans. Audio Speech Lang. Process.* **32**, 3618–3630. (2024).
30. Liu, Q., Tong, W., Bai, H. & Dong, S. SCBM-Net: a multimodal feature fusion-based dual-channel method for bearing fault diagnosis. *Sci. Rep.* **15** (1), 37904 (2025).
31. Dong, S., Tong, W., Bai, H. & Liu, Q. Bearing fault diagnosis method based on WSST and ISSA-MCNN-BIGRU. *Sci. Rep.* **15** (1), 41533 (2025).
32. Chen, Z. et al. Learning multi-scale features for speech emotion recognition with connection attention mechanism. *Expert Syst. Appl.* **214**, 118943 (2023).
33. Ivanko, D., Ryumin, D. & Karpov, A. A review of recent advances on deep learning methods for audio-visual speech recognition. *Mathematics* **11** (12), 2665 (2023).
34. Prashanth, A., Jayalakshmi, S. L. & Vedhapriyavadhana, R. A review of deep learning techniques in audio event recognition (AER) applications. *Multimedia Tools Appl.* **83** (3), 8129–8143 (2024).
35. He, Y., Seng, K. P. & Ang, L. M. Multimodal Sensor-Input Architecture with Deep Learning for Audio-Visual Speech Recognition in Wild. *Sensors* **23** (4), 1834 (2023).
36. Sun, J., Gao, P., Liu, P. & Wang, Y. Memristor-Based Feature Recall Neural Network Circuit With Temporal Differentiation of Emotion and its Application in Parts Inspection. *IEEE Trans. Industr. Inf.* **21** (7), 5633–5643 (2025).
37. Sun, J., Gao, P., Wen, S., Liu, P. & Wang, Y. Memristor-based conditioned inhibition neural network circuit with blocking generalization and differentiation. *IEEE Internet Things J.* **11** (7), 11259–11270 (2023).

Acknowledgments

This work was supported by Research Project on the Protection and Contemporary Value of Heilongjiang Manchurian Music, 2025 Basic Research Business Expenses of Provincial Universities in Heilongjiang Province.

Author contributions

Chang Liu: Conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation. Ningning Shi: methodology, software, validation, formal analysis. Shen Jiang: writing—review and editing, visualization, supervision, project administration, funding acquisition.

Declarations

Competing interests

The authors declare no competing interests.

Ethics statement

The studies involving human participants were reviewed and approved by College of Arts, Heilongjiang University Ethics Committee (Approval Number: 2021.398472). The participants provided their written informed consent to participate in this study. All methods were performed in accordance with relevant guidelines and regulations.

Additional information

Correspondence and requests for materials should be addressed to N.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026